

BayLearn Adaptive Learning Platform

Module 2: Question Generation & Assessment System

Updated Architectural Design — v3.0

Incorporates Expert Question Bank · Bloom Difficulty Calibration · Rejection Feedback · Multi-Modal Support · Student Explanation Elicitation

Graduation Project · Computer Science & AI · 2024–2025

2-Student Team | Parallel Development Track

0. Change Log — v3.0 Updates

The following additions and changes have been made to the plan in this revision:

Change	Type	Priority	Location
Expert Question Bank (Section 4.0)	New Section — Foundational Infrastructure	★ HIGH	Sec. 4.0
Bloom Difficulty LLM Judge	New Component — Assessment Engine	★ HIGH	Sec. 5.3
Question Rejection Feedback Loop	New Component — Verifier Gauntlet	★ HIGH	Sec. 3.4
Multi-Modal Question Support	New Feature — STEM Enhancement	◆ MEDIUM	Sec. 3.5
Student Explanation Elicitation	New Feature — Assessment Engine	◆ MEDIUM	Sec. 5.4
Collaborative/Group Question Modes	Deferred — Low Priority	⚠ LOW	Future Work
Teacher Dashboard & Override UI	Deferred — Low Priority	⚠ LOW	Future Work
Cross-Session Difficulty Adaptation	Deferred — Low Priority	⚠ LOW	Future Work
2-Student Work Division (Section 10)	New Section — Project Management	REQUIRE D	Sec. 10
Full Upgrade Pipeline Update (Section 9)	Revised — Added L0, updated L1–L4	REQUIRE D	Sec. 9

1. Module Overview & Objectives

Module 2 is the intelligence core of BayLearn, responsible for generating pedagogically grounded questions and performing multi-dimensional student assessment. It interfaces upstream with student study materials (via the RAG pipeline) and downstream with the Reinforcement Learning adaptive engine (EPPO + simpleKT) via a defined API contract.

1.1 Core Responsibilities

- Generate questions grounded in student-supplied study material
- Align questions to Bloom's Taxonomy cognitive levels (L1–L6)
- Calibrate difficulty via Item Response Theory (IRT / BERT-IRT)
- Verify question quality through a multi-stage gauntlet
- Grade student responses and return partial credit + confidence scores
- Return enriched RL signals to EPPO for adaptive learning decisions
- **NEW: Ground generation in an Expert Question Bank for pedagogical accuracy**

1.2 Module Pillars

Pillar	Description	Impact
Pillar 1 — Question Quality	Multi-stage verification, CoVe hallucination reduction, expert grounding	Ensures questions are accurate, diverse, well-calibrated
Pillar 2 — Sensor Bandwidth	Rich multi-dimensional signals beyond binary correct/incorrect (explanation elicitation, partial credit, confidence)	Gives EPPO more to learn from per interaction
Pillar 3 — Adaptive Targeting	IRT-based CAT with information-gain selection, online calibration	Minimizes questions needed to accurately estimate student ability

2. System Architecture Overview

The module is structured into four functional layers, executed in sequence for each question-generation event:

Layer 0 — Foundation

Expert Question Bank (infrastructure) — provides few-shot demonstrations indexed by concept × Bloom level

Layer 1 — Question Generation

RAG Context Retriever → Expert Question Retriever → Generator Ensemble (CoVe / NSDG / LLM-Modulo) → Verifier Gauntlet (with Rejection Feedback) → Multi-Modal Renderer

Layer 2 — Assessment

IRT Difficulty Calibrator (BERT-IRT) → Bloom LLM Judge → LLM Grader → Partial Credit Engine → Explanation Elicitor → Enriched Signal Composer

Layer 3 — Adaptive Testing

CAT Engine (Information Gain Selection) → EPPO API Contract → RL Signal Delivery

3. Question Generation Pipeline

3.1 Context Retriever (RAG)

The retrieval pipeline encodes student study material using a bi-encoder (e.g., sentence-transformers/all-mpnet-base-v2) and retrieves semantically relevant passages using FAISS or ChromaDB as the vector store. Retrieval is concept-aware: queries are formed using the target concept tag to ensure retrieved context is topically aligned.

Component	Technology	Output
Encoder	sentence-transformers/all-mpnet-bas e-v2	Dense passage vectors (768-dim)
Vector Store	FAISS (local) / ChromaDB (scalable)	Top-k retrieved passages
Query Construction	Concept + Bloom level template	Targeted semantic query
Passage Scoring	Cosine similarity + recency weighting	Ranked passage list

3.2 Generator Ensemble

Three complementary generation strategies are used in ensemble to maximize question diversity and quality:

3.2.1 CoVe — Chain-of-Verification

Reduces hallucination by generating verification questions about each claim in the drafted question, then self-checking answers. Questions that fail verification are flagged and regenerated. This is the primary hallucination-reduction mechanism.

3.2.2 NSDG — Non-Sampling Diverse Generation

Generates a diverse set of question candidates by steering the decoding process to maximize diversity in the output space. Each candidate targets a different pedagogical angle on the same concept. Prevents repetitive question formats.

3.2.3 LLM-Modulo

A planning-as-satisfiability approach that treats question generation as a constrained optimization: the LLM proposes candidates, external critics (Bloom classifier, groundedness checker, difficulty estimator) evaluate them, and the LLM revises until all critics are satisfied. Ensures structural compliance before the Verifier Gauntlet is even reached.

3.3 Bloom Taxonomy Prompt Engineering

Each generation call is parameterised with a Bloom level (1–6) that controls the cognitive demand of the generated question. The prompt includes:

- The target Bloom level label (e.g., 'Analysis — Bloom L4')
- The operational verbs for that level (e.g., 'compare, differentiate, examine')
- **Two few-shot demonstrations retrieved from the Expert Question Bank at the same level**
- The relevant study material passages from the RAG retriever

3.4 Verifier Gauntlet

Every generated question passes through a sequential verification pipeline before being accepted:

Stage	Verifier	Method	Rejection Rate Target
V1	Groundedness Checker	NLI model — question claims must be entailed by source passages	<15%
V2	Bloom Level Classifier	Fine-tuned BERT classifier — verifies question matches target cognitive level	<10%
V3	Answerability Checker	LLM self-answer: if the model cannot answer the question from the passage, reject	<8%
V4	Distractor Quality	Checks each distractor is plausible but unambiguously wrong	<12%
V5	Duplicate Detector	Semantic deduplication against session question pool (cosine threshold 0.85)	<5%

★ NEW — HIGH IMPACT / LOW EFFORT

Question Rejection Feedback Loop

- When any verifier rejects a question, the rejection reason is serialised and injected back into the regeneration prompt as an explicit constraint.
- Example: 'REJECTION REASON: V1 Groundedness — the question referenced a formula not present in the source passages. Regenerate without referencing unsupported claims.'
- This targeted feedback reduces retry budget exhaustion significantly compared to blind regeneration.
- Implementation: Each Verifier returns a structured rejection object {stage, reason, constraint_hint} consumed by the Prompt Builder.
- Retry budget: Maximum 3 regeneration attempts before escalating to a fallback template-based question.

3.5 Error Taxonomy Distractor Engine

Distractors are not randomly sampled — they are generated to represent specific student error patterns identified in the literature:

- Conceptual confusion: plausible but category-wrong answers (e.g., wrong layer in networking)
- Off-by-one/boundary errors: common in algorithmic and mathematical questions
- Partial knowledge: answers correct for a simpler version of the concept
- Terminology conflation: answers that mix up related but distinct terms

◆ NEW — MEDIUM IMPACT / MEDIUM EFFORT

Multi-Modal Question Support (STEM)

- For CS/engineering subjects, questions may embed a code snippet, pseudocode block, circuit diagram (ASCII or SVG), or a data structure diagram as part of the question stem.
- Implementation: the Generator Ensemble has a modality flag — text (default), code, diagram. The Prompt Builder selects a modality-appropriate template.
- Code snippets are syntax-highlighted in the student UI. Pseudocode uses a standard notation agreed between both students.
- This is a strong differentiator for a STEM-focused adaptive learning platform.
- Complexity management: for graduation scope, implement text + code modalities. Diagram generation (SVG) is deferred to post-graduation enhancement.

4. Expert Question Bank — Foundational Infrastructure

★ NEW SECTION — Foundational to all higher layers

The Expert Question Bank is a vector-indexed repository of professionally authored questions tagged with structured metadata. It serves as the epistemic grounding layer for the entire generation pipeline — rather than generating questions blindly from a system prompt, the LLM generates in the style of a human expert who wrote real exam questions.

4.1 Purpose and Rationale

Motivation (from literature): Fabbri et al. (2021) showed that template retrieval improves question diversity and difficulty calibration. Kulshreshtha et al. (2021) demonstrated that few-shot QG using retrieved similar questions significantly outperforms zero-shot generation. Wang et al. (2022, EduQG) explicitly used retrieved textbook examples as generation demonstrations.

The specific angle of indexing by Bloom level and using it as a retrieval dimension is underexplored — this is a genuine novelty contribution claimed by this project.

4.2 Question Bank Schema

Field	Type	Description	Example
question_id	UUID	Unique identifier	qb-00a1...
question_text	String	The question stem (may include code block)	Explain the TCP 3-way handshake...
bloom_level	Int 1–6	Bloom taxonomy cognitive level	4 (Analysis)
concept_tags	String[]	Topical tags from a controlled vocabulary	['TCP', 'networking', 'handshake']
subject_domain	Enum	High-level domain	CS / Engineering / Math
difficulty_irt	Float	Pre-calibrated IRT difficulty (β)	1.24
reference_answer	String	Model answer for grading anchor	The client sends SYN...
source	String	Provenance of question	MIT OCW 6.829 Pset 3
embedding	Float[768]	Dense vector for semantic retrieval	(stored in vector DB)

4.3 Data Sources

- MIT OpenCourseWare problem sets (CS, EE, Math)
- ARC (AI2 Reasoning Challenge) — structured reasoning questions

- MMLU (Massive Multitask Language Understanding) — 57-domain academic questions
- SciQ — science exam questions with answer explanations
- ExamQA — diverse exam questions across subjects
- Khan Academy exercises (via public API / scraping within ToS)
- CS/engineering textbook end-of-chapter problems (manually curated)

4.4 Few-Shot Retrieval Integration

At generation time, the Expert Question Retriever is called in parallel with the RAG Context Retriever:

Step	Action	Output
1	Query: {concept: 'TCP handshake', bloom_level: 4}	Structured vector query
2	Retrieve Top-2 questions from bank matching concept + bloom_level	2 expert question objects
3	Include question_text + reference_answer in generation prompt as demonstrations	Prompt enriched with expert examples
4	LLM generates 'in the style of' the expert examples	Pedagogically grounded question

4.5 Reference Answer Grounding

When a reference answer exists in the Expert Question Bank for a similar concept, it is passed to the Assessment Engine as a grounding anchor. The LLM Judge compares the student's answer against the reference — not just its own generated rubric — making grading significantly more reliable.

4.6 Computational Requirements

Component	Compute Cost	Storage	Complexity
Few-shot retrieval from bank	Very Low (+~500 tokens/request)	~1 GB for 100k questions	Low — slots into existing RAG
Reference answer grounding	Near Zero (part of Assessment prompt)	Part of bank schema	Low
Initial bank population & embedding	One-time batch job, ~4 GPU-hours	As above	Low-Medium

5. Assessment Engine

5.1 IRT-Based Difficulty Calibration (BERT-IRT)

Item Response Theory (IRT) models the relationship between student ability (θ) and question difficulty (β) using a logistic function. We implement BERT-IRT: a neural extension that computes difficulty from question text encoding, enabling difficulty estimation for new (unseen) questions without requiring prior response data.

The 2-Parameter Logistic (2PL) IRT model:

$$P(\text{correct} \mid \theta, \alpha, \beta) = 1 / (1 + \exp(-\alpha(\theta - \beta)))$$

- θ = latent student ability (estimated by simpleKT / updated by EPPO)
- β = question difficulty (estimated by BERT-IRT from question text)
- α = discrimination parameter (how well the question separates ability levels)

5.2 LLM Judge Grading

An LLM-as-judge evaluates student free-text responses against the question and reference answer.

The judge returns:

- Correctness score (0–1, continuous)
- Partial credit breakdowns by sub-criterion (for multi-part questions)
- Misconception tags: identifies which specific error the student made
- Confidence level of the grading decision (used for downstream uncertainty weighting)

★ NEW — HIGH IMPACT / LOW EFFORT

Bloom Difficulty Calibration via LLM Judge

- Before a question is sent to the student, a dedicated LLM judge rates its perceived difficulty on a 1–5 Likert scale, independently of its Bloom taxonomy label.
- Rationale: Bloom is a cognitive category, NOT a difficulty score. A Bloom-2 question can be very hard if the concept is advanced. Having both dimensions gives IRT cleaner signal.
- Output: each question carries two independent scores — bloom_level (cognitive type) and perceived_difficulty (1–5 Likert from LLM judge).
- Both are stored in the question metadata and passed to EPPO for richer adaptive decisions.
- Implementation: a single additional LLM call at question generation time (~200 tokens). Very low cost.

5.3 Partial Credit Engine

Binary correct/incorrect grading discards learning signal. The Partial Credit Engine awards fractional scores based on:

- Conceptual correctness of each sub-component of the answer
- Evidence of reasoning process even when final answer is wrong
- Language / precision of explanation

◆ NEW — MEDIUM IMPACT / MEDIUM EFFORT

Student Explanation Elicitation

- On a configurable fraction of questions (default: 30%), after the student submits an answer, a follow-up prompt is shown: 'Briefly explain your reasoning in one or two sentences.'
- The explanation is fed into the LLM Judge as additional evidence for grading.
- Critical capability: explanations catch confident-but-wrong students who guessed correctly — binary correct/incorrect and even partial credit miss this pattern entirely.
- The explanation quality score (0–1) is included in the enriched RL signal sent to EPPO, giving a richer estimate of conceptual mastery.

- Student experience: framing as 'help us understand your thinking' rather than 'explain yourself' improves compliance. Explanation is optional (skip allowed) but contributes to score.

6. Computerized Adaptive Testing (CAT)

6.1 Information Gain Selection

Rather than selecting questions at random or in fixed sequence, the CAT engine selects the question that maximises Fisher Information at the current estimated student ability θ . This minimises the number of questions needed to achieve a target measurement precision.

6.2 Item Bank Management

The CAT draws from the Verified Question Pool — questions that have passed the Verifier Gauntlet and have calibrated IRT parameters. The pool is partitioned by subject domain and concept, allowing the CAT to enforce coverage constraints (ensuring all major topics are visited within a session).

6.3 Termination Criteria

- Standard error of θ estimate falls below threshold ($SE < 0.3$)
- Maximum session question count reached (configurable, default 20)
- All target concepts have been covered at least once

7. Enriched RL Signal — Pillar 2 (Sensor Bandwidth)

The standard adaptive learning loop returns only binary correct/incorrect to EPPO. Module 2's core research contribution (Pillar 2) is to dramatically expand this bandwidth:

Signal Component	Source	What EPPO Gains
Correctness score	LLM Judge (0–1 continuous)	Granular mastery estimate
Bloom level of question answered	Generator + Bloom Classifier	Knowledge of which cognitive skill was tested
Perceived difficulty (Likert 1–5)	NEW: Bloom LLM Judge	True difficulty signal independent of taxonomy level
Misconception tag(s)	LLM Judge error taxonomy	Specific gap identification for curriculum routing
Explanation quality score	NEW: Explanation Elicitor (0–1)	Confidence vs. guessing signal
Response time (normalised)	Frontend timestamp	Confidence proxy — fast+wrong differs from slow+wrong

Signal Component	Source	What EPPO Gains
IRT parameters (α , β)	BERT-IRT	Allows EPPO to model question characteristics explicitly

7.1 API Contract with EPPO (simpleKT)

The following JSON structure is emitted by Module 2 to EPPO after each question response:

```
{  "student_id": "str",  "session_id": "str",  "question_id": "str",  "concept": "str",  "bloom_level": "int (1-6)",  "perceived_difficulty": "float (1-5)",  "irt_beta": "float",  "irt_alpha": "float",  "correctness": "float (0-1)",  "partial_credit_breakdown": "float[]",  "misconception_tags": "str[]",  "explanation_quality": "float | null",  "response_time_ms": "int",  "timestamp": "ISO-8601"}
```

8. Deferred Features — Low Priority

The following features were considered in the design but are explicitly deferred. They should be mentioned as Future Work in the evaluation chapter. Do not implement for the graduation submission.

△ LOW PRIORITY Collaborative / Group Question Modes

- Allow multiple students to answer a shared question set simultaneously
- Requires real-time session synchronization infrastructure
- Significant frontend and backend complexity beyond graduation scope

Recommended disposition: Mention as Future Work in evaluation chapter.

△ LOW PRIORITY Teacher Dashboard & Override Interface

- Allows instructors to review generated questions, override difficulty ratings, flag incorrect questions
- Useful for deployment but not required to validate the research contributions
- Builds on Module 2 outputs — can be added post-graduation

Recommended disposition: Mention as Future Work in evaluation chapter.

△ LOW PRIORITY Cross-Session Difficulty Adaptation

- Persisting and updating IRT θ estimates across multiple learning sessions
- Requires persistent student model storage and integration with simpleKT long-term memory
- The EPPO team's responsibility — Module 2 emits per-session signals only

Recommended disposition: Mention as Future Work in evaluation chapter.

△ LOW PRIORITY Full Online IRT Calibration (L4)

- Online Expectation-Maximization updating β parameters as new student responses arrive
- High algorithmic complexity — implement a lite version (EMA smoothing) for graduation, mention full EM as future work
- Can be demonstrated in simulation without actual large-scale response data

Recommended disposition: Mention as Future Work in evaluation chapter.

9. Full Upgrade Pipeline — MVP → L0 → L1 → L2 → L3 → L4

This section describes the staged development plan. Each layer is a superset of the previous. The layers correspond to implementation milestones, not separate systems.

Layer	Name	Key Additions	Priority	Dependency
MVP	Minimum Viable Pipeline	Basic RAG retrieval + LLM generation (zero-shot) + keyword grading + binary correct/incorrect output to EPPO	MUST HAVE	None
L0 ★ NEW	Expert Knowledge Foundation	Expert Question Bank (populated from ARC/MMLU/MIT OCW) + Few-shot retrieval at generation time + Reference answer grounding in Assessment	MUST HAVE	MVP
L1	Quality Guarantees	Full Verifier Gauntlet (V1–V5) + CoVe hallucination reduction + Rejection Feedback Loop (NEW) + Error Taxonomy Distractor Engine	MUST HAVE	L0
L2	Rich Signal	Bloom LLM Judge (NEW — perceived difficulty) + Enriched RL Signal composition + Student Explanation Elicitation (MEDIUM)	SHOULD HAVE	L1
L3	Adaptive Targeting	BERT-IRT calibration + CAT information-gain selection + Multi-Modal questions for STEM (MEDIUM)	SHOULD HAVE	L2
L4 △ LOW	Online Learning	Online IRT calibration (EM) + Deduplication across sessions + Cross-session adaptation	FUTURE WORK	L3

9.1 Graduation Delivery Scope

For the graduation submission, the target delivery is:

Required: MVP + L0 + L1 (fully implemented and evaluated)

Target: L2 fully implemented + L3 partially implemented (IRT calibration + basic CAT)

Stretch: L3 fully implemented (including Multi-Modal support)

Defer: L4 — mention in evaluation chapter as future work only

10. 2-Student Work Division — Black Box Architecture

The module is divided into two fully independent subsystems. Each student develops their subsystem as a black box with formally specified inputs and outputs. The division is designed so that neither student has an implementation dependency on the other — they can develop, test, and demo in parallel.

STUDENT A — Question Generation & Knowledge Infrastructure

10.1.A Responsibilities

- Expert Question Bank: design schema, source datasets, populate, embed, and index
- RAG Context Retriever: encoder, vector store, passage scoring
- Expert Question Retriever: concept×bloom_level retrieval from bank
- Generator Ensemble: CoVe, NSDG, LLM-Modulo implementations
- Bloom Prompt Engineering: level-parameterised templates + few-shot injection
- Verifier Gauntlet: V1–V5 verifiers + Rejection Feedback Loop
- Error Taxonomy Distractor Engine
- (Medium Priority) Multi-Modal Question Renderer: code/pseudocode support

10.2.A Assumed Input (Black Box Entry Point)

```
{ "student_id": "str", "session_id": "str", "concept": "str", // e.g.
  "TCP handshake" "bloom_level": "int (1-6)", // requested cognitive level
  "study_material": "str", // raw text of student's uploaded material
  "session_question_pool": "str[]" // IDs of questions already asked this session}
```

10.3.A Assumed Output (Black Box Exit Point)

```
{ "question_id": "str", "question_text": "str", "modality": "text | code | diagram",
  "bloom_level": "int (1-6)", "concept_tags": "str[]", "distractors": ["str", "str",
  "str"], // MCQ options (wrong answers) "correct_answer": "str", "reference_answer":
  "str", // full model answer for grading anchor "difficulty_irt_estimate":
  "float", // initial BERT-IRT estimate from question text "expert_bank_grounding":
  "bool", // was a bank example used? "generation_metadata": {
  "verifier_passes": "str[]", "retries": "int", "model_used": "str" }}
```

10.4.A Testing Without Student B

- Use a mock Assessment Engine that returns random scores from a fixed distribution
- Validate: question groundedness (V1), Bloom level accuracy (V2), distractor plausibility
- Benchmark: measure rejection rates at each Verifier stage, question diversity (embedding spread)
- Evaluate few-shot vs. zero-shot generation quality using human ratings on a 20-question sample

STUDENT B — Assessment Engine, CAT & RL Signal

10.1.B Responsibilities

- IRT Difficulty Calibrator: BERT-IRT implementation and θ estimation
- Bloom Difficulty LLM Judge: Likert 1–5 perceived difficulty rating (NEW)
- LLM Grader: evaluates student responses against reference answers
- Partial Credit Engine: sub-criterion scoring
- (Medium Priority) Student Explanation Elicitor: follow-up prompt + explanation scoring

- CAT Engine: information-gain question selection from verified pool
- Enriched RL Signal Composer: builds complete signal object for EPPO
- EPPO API Interface: serialises and delivers signal to the RL engine

10.2.B Assumed Input (Black Box Entry Point)

```
// QUESTION OBJECT (received from Student A's output):{ "question_id": "str",
"question_text": "str", "bloom_level": "int (1-6)", "concept_tags": "str[]",
"correct_answer": "str", "reference_answer": "str", "difficulty_irt_estimate": "float",
"distractors": "str[]"}// STUDENT RESPONSE (received from frontend):{ "student_id":
"str", "session_id": "str", "question_id": "str", "student_answer": "str",
"student_explanation": "str | null", // may be null if elicitation skipped
"response_time_ms": "int", "current_theta_estimate": "float" // from simpleKT /
EPPO}
```

10.3.B Assumed Output (Black Box Exit Point)

```
// ENRICHED RL SIGNAL (emitted to EPPO after each response):{ "student_id": "str",
"session_id": "str", "question_id": "str", "concept": "str", "bloom_level": "int
(1-6)", "perceived_difficulty": "float (1-5)", // from Bloom LLM Judge "irt_beta":
"float", // calibrated difficulty "irt_alpha": "float",
// discrimination "correctness": "float (0-1)", "partial_credit_breakdown": "float[]",
"misconception_tags": "str[]", "explanation_quality": "float | null",
"response_time_ms": "int", "updated_theta": "float", // revised ability
estimate "next_question_hint": { "target_bloom": "int", "target_difficulty":
"float", "target_concept": "str | null" }, "timestamp": "ISO-8601"}
```

10.4.B Testing Without Student A

- Use a mock Question Generator that samples questions from the Expert Question Bank directly (this is valid because the bank exists as standalone infrastructure)
- Validate: IRT parameter stability across simulated response sequences, misconception tag accuracy
- Simulate student ability levels (low/medium/high) and verify CAT converges faster than random selection
- Benchmark: measure θ estimation accuracy against known simulated ability using MSE

10.5 Integration Protocol

The two subsystems are integrated via a simple REST interface. Student A exposes a /generate endpoint; Student B exposes a /assess endpoint. The shared contract is the JSON schemas defined in sections 10.3.A and 10.2.B above.

Integration Point	Owner	Protocol	Validation Method
Question Generation API	Student A	REST POST /generate → returns Question Object	Schema validation (JSON Schema)
Assessment API	Student B	REST POST /assess → returns RL Signal	Schema validation + EPPO sandbox

Integration Point	Owner	Protocol	Validation Method
Expert Question Bank	Student A owns, Student B reads	Direct DB read (read-only for Student B)	Shared test fixture

11. Evaluation Framework

11.1 Question Quality Metrics (Student A)

Metric	Method	Target
Groundedness	NLI entailment score vs. source passages	> 0.85 average
Bloom Level Accuracy	Expert-annotated test set (n=100)	> 80% correct classification
Few-Shot vs. Zero-Shot Quality	Human rating (1–5 Likert, n=20 questions)	>0.5 point improvement
Distractor Plausibility	Human judges — are wrong answers tempting?	> 70% rated 'plausible'
Verifier Pass Rate	Per-stage rejection statistics	Overall < 30% total rejection

11.2 Assessment Quality Metrics (Student B)

Metric	Method	Target
Grading Accuracy	Comparison to human graders on 50-response set	> 85% agreement
IRT Calibration	θ estimation MSE on simulated cohort	MSE < 0.2
CAT Efficiency	Questions to convergence vs. random selection	> 25% reduction
Signal Richness	Correlation of RL signal components with learning outcomes	> 0.5 correlation

12. Innovation Layer — Research Contributions

Contribution	Novelty Claim	Supporting Literature
Expert Question Bank with Bloom-level retrieval	Indexing by Bloom taxonomy for few-shot retrieval is underexplored — genuine novelty	Fabbri (2021), Kulshreshtha (2021), EduQG Wang (2022)
Rejection Feedback Loop in Verifier Gauntlet	Targeted constraint propagation from verifiers to generator — novel combination	LLM-Modulo (Kambhampati 2024)

Contribution	Novelty Claim	Supporting Literature
Bloom LLM Judge — perceived difficulty vs. taxonomy level	Separation of cognitive category from difficulty score in IRT — novel pipeline component	IRT literature + Bloom taxonomy research
Explanation Elicitation → RL signal	Explanation quality as a component of adaptive learning reward — not seen in prior work	Pillar 2 — Sensor Bandwidth thesis
Enriched Multi-Dimensional RL Signal	7-component signal vs. binary correct/incorrect — core Pillar 2 contribution	EPPO + simpleKT (Content-Aware)